

Accountable-Authority Identity-Based Matchmaking Encryption and Its Application

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Abstract—Identity-based matchmaking encryption (IB-ME) is the identity-based instantiation of matchmaking encryption, which provides identity privacy, bilateral access control, confidentiality and authenticity of messages. Due to its compelling features, IB-ME is a promising technique that provides privacy-enhanced bilateral access control. However, IB-ME confronts the problems of key escrow and accountability for malicious behaviors since the authority can generate the encryption key and decryption key for any identity. To mitigate the above issues, we propose the concept of accountable-authority IB-ME (A-IB-ME) and present its first formal definition and concrete construction, which supports the accountability of malicious behaviors and achieves the functionalities of IB-ME. In A-IB-ME, when an encipher box and a decoder box are created using respective keys, the creator can be implicated if necessary, enabling accountability for malicious behavior of key abuse in the event of disputes arising on responsibility. Then, we extend the security model of A-IBE to the A-IB-ME scenario and present the formal proof under the standard assumptions without random oracles. Next, we analyze and evaluate the performance of A-IB-ME theoretically and experimentally, demonstrating its efficiency. Finally, we discuss the applications of A-IB-ME and present an instance by applying it to privacy-preserving user matching in mobile social networks.

Index Terms—Identity-based matchmaking encryption, accountable, key escrow, malicious behaviors, privacy-preserving user matching.

I. INTRODUCTION

MATCHMAKING Encryption (ME) [1] brings a novel secret communication method, which not only

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simultaneously provides the privacy and authenticity of messages as well as the attribute privacy, but also provides bilateral fine-grained access control over encrypted data. ME requires the sender and receiver to specify the access control policies satisfied by the other party simultaneously to decrypt the ciphertext. When decryption fails, the receiver does not learn which access control policy is not satisfied. Besides, ME does not require both parties to be online at the same time, naturally preventing traffic analysis attacks. At CRYPTO 2019 [1], Ateniese et al. instantiated the concept of ME in the identity-based setting (i.e., identity-based ME (IB-ME)). In IB-ME, the sender and receiver obtain the encryption key and decryption key from the authority according to their identities, respectively. When encrypting a message, the sender needs to input the encryption key and the intended receiver's identity. When decrypting a ciphertext, the receiver needs to input the decryption key and the target sender's identity. The ciphertext can be decrypted correctly only when the identity of the sender or receiver is identical to the identity specified by the other party. Due to its promising features, IB-ME has garnered significant interest from researchers. However, existing IB-ME schemes [2], [3] require further enrichment and improvement.

A motivational story: IB-ME offers enhanced privacy and security that can be applied to various scenarios, including anonymous bulletin boards, social networks, and other bilateral access control. In these scenarios, security and privacy depend on the trust in the authority, who generates the encryption key and decryption key for any users with different identities. However, full trust in the authority may lead to the inherent key escrow problem [4] and accountability for malicious behaviors [5]. Consider the following scenarios in social networks where an anonymous service platform is utilized by the sender and receiver to transmit information. In the process, the platform can provide bilateral access control, identity privacy of the sender and receiver, as well as message confidentiality and authenticity. When the platform discovers that an encipher box from the user Alice is maliciously propagated, Alice may face the risks of being punished. But Alice claims to be innocent and accuses the platform of framing her. How can her innocence or guilt be established? Can Alice present a defense for herself? Similarly, if the platform unveils that a decoder box belonging to Bob is unrightfully shared on the platform and Bob denies any wrongdoing, what actions can we follow to prove that Bob is not responsible, and will he have the opportunity to contest the allegation against him? On the discussed platform, both the users

(in this case, Alice and Bob) and the platform can generate the encipher and decoder boxes. In the event of disputes arising on responsibility, it is not possible to hold anyone accountable for key abuse without concrete evidence. Furthermore, the authority may intercept communication data between users without consent, raising concerns about breaches of privacy. Additionally, users need assurances that the authority does not issue their keys to potentially malicious individuals. If an encryption or decryption key is misused or abused [4], [6], [7], the current IB-ME lacks the ability to assign blame since both the user and authority possess the capacity to do so.

In identity-based encryption (IBE), to mitigate the problems of key escrow and key abuse, Goyal [8] proposed the concept of accountable authority IBE (A-IBE) by reducing the trust in the authority. Similarly, a very natural idea is to introduce the concept and construction of accountable-authority IB-ME (A-IB-ME) to mitigate the problems of key escrow and trace the source of malicious behaviors when key abuses happen in IB-ME. Although some researchers [2], [3], [9] have improved the definition of IB-ME and enriched its application scenarios, there is no study considering the accountable-authority setting. This could be attributed to the fact that implementing A-IB-ME is not a straightforward task. One simple approach to reduce the trust in the authority is to leverage multi-authority schemes [10], [11]. Nonetheless, this solution increases the costs of communication and infrastructure and may cause security issues of collusion among multiple authorities. Additionally, there are some accountable-authority schemes [5], [12], [13], [14] in IBE. However, these schemes cannot be extended directly. On one hand, they do not consider identity privacy. In A-IBE [5], it is necessary to check the sanity of ciphertexts, leading to loss of anonymity, an important property in A-IB-ME. Moreover, additional functionalities of bilateral access control and message authenticity also increase the difficulty of preserving identity privacy. On the other hand, in A-IB-ME, malicious behaviors are classified into sender-related and receiver-related types while only the latter is taken into account in A-IBE. In this case, the authority's malicious behavior should be considered in two groups respectively as well. New syntax and security model need to be defined to capture these behaviors. To solve the accountability concern of IB-ME, we formalize the first A-IB-ME and present a concrete scheme.

A. Our Contributions

Our work has the following contributions:

- Firstly, we introduce the concept of A-IB-ME and present the formal syntax and security definitions, extending and enriching the existing IB-ME framework. In A-IB-ME, when the encryption key and decryption key are abused to create an encipher box and decoder box, the tracing algorithm can implicate its creator. Compared with IB-ME, A-IB-ME weakens the trust in the authority, mitigates the key escrow, and achieves accountability for malicious behaviors.
- Secondly, we propose a concrete scheme of A-IB-ME and prove its privacy, authenticity, dishonest-sender security,

dishonest-receiver security, and dishonest-authority security, which are reduced to the decisional bilinear diffie-hellman (DBDH), decision linear (D-Lin), computational bilinear diffie-hellman (CBDH) assumptions without random oracles, where our construction is based on static assumptions and does not require other cryptographic tools such as non-interactive zero-knowledge proof systems.

- Finally, we evaluate its performance theoretically. To show the efficiency of A-IB-ME more intuitively, we implement A-IB-ME. Theoretical and experimental results demonstrate the efficiency of A-IB-ME. Also, we provide an application instance that shows how A-IB-ME offers a privacy-enhanced user matching mechanism in mobile social networks.

To more distinctly accentuate the distinctive features of our scheme, we carry out a detailed comparison with existing cryptographic primitives and relevant schemes. The specifics of the comparison are presented in Table I.

Differences from IB-ME and A-IBE: A-IB-ME offers more functionalities compared to the other two cryptographic primitives (i.e., IB-ME and A-IBE).

A-IB-ME addresses the issues of key escrow and malicious behavior accountability in IB-ME, which have not been addressed by the existing extensions like Fuzzy IB-ME and IB-BME. The main differences between A-IB-ME and IB-ME are as follows:

- *Syntax Definition:* A-IB-ME provides additional algorithms to check the encryption key and decryption key format integrity and identify the initiator of malicious behavior in case of liability disputes.
- *Security Definition:* A-IB-ME not only ensures confidentiality and authenticity of messages, but it also guarantees dishonest-sender security, dishonest-receiver security, and dishonest-authority security.

As for the differences between A-IB-ME and A-IBE, A-IB-ME goes beyond by providing bilateral access control, message authenticity, and identity privacy. Consequently, the two cryptographic primitives differ in syntax and security definitions.

- *Syntax Definition:* A-IB-ME has added an encryption key generation algorithm, which is used in the process of encrypting messages by the sender. This also enables A-IB-ME to provide the message authenticity. In the decryption phase, the receiver also needs to input the target sender's identity. If a receiver is not the intended receiver of the ciphertext, the receiver cannot decrypt the ciphertext regardless of the identity of any sender entered. In addition, for non-intended receivers, the sender's input (i.e. message, sender identity, and receiver identity information) is private.
- *Security Definition:* A-IB-ME needs to provide the message authenticity, identity privacy, dishonest-sender security, while A-IBE cannot provide such security.

Differences from the existing IB-ME schemes against key escrow issues: While several existing IB-ME schemes [9], [17], [18] offer solutions to key escrow problems, they differ significantly from our scheme in terms of functionality, syntax definition, and security protocols. Notably, our scheme incorporates

TABLE I
FEATURE COMPARISON WITH RELATED SCHEMES

Schemes	Anonymity	Confidentiality	Authenticity	Bilateral Access Control	Key Escrow	Accountability	Models
A-IBE [5]	×	✓	×	×	✓	✓	Standard Model
IB-ME [1]	✓	✓	✓	✓	×	×	Random Model
IB-ME [2]	✓	✓	✓	✓	×	×	Plain Model
IB-ME [3]	✓	✓	✓	✓	×	×	Standard Model
Fuzzy IB-ME [15]	✓	✓	✓	✓	×	×	Standard Model
IB-BME [16]	✓	✓	✓	✓	×	×	Standard Model
CL-ME [9]	✓	✓	✓	✓	✓	×	Random Model
RME [17]	✓	✓	×	✓	✓	×	Plain Model
BIB-ME [16]	✓	✓	✓	✓	✓	×	Random Model
Ours	✓	✓	✓	✓	✓	✓	Standard Model

Anonymity means that identity information does not be leaked; Confidentiality represents message privacy; Authenticity means that the identity of the encryptor is unforgeable; Key escrow refers to the ability of a designed scheme to resist key escrow issues; Accountability means that key abuse can be held accountable.

accountability mechanisms to address key abuse, a feature that other schemes [9], [17], [18] have not adequately resolved. The distinctions in other areas are outlined below:

- *Certificateless Matchmaking Encryption (CL-ME)* [9]: The proposed scheme exhibits distinct differences from CL-ME concerning syntax and security definitions. In particular, CL-ML locally generates user public keys as part of the encryption and decryption key generation process. Within the security framework, adversaries have the potential to initiate public key substitution attacks; however, this is not a security consideration in the proposed scheme.
- *Registered Matchmaking Encryption (RME)* [17]: Our proposed scheme stands apart from RME in terms of both functionality and definition. RME falls short in providing message authenticity functionality, whereas our scheme successfully incorporates this capability. This divergence consequently results in notable differences in the syntax and security definitions between our scheme and RME. Furthermore, the implementation of RME hinges on intricate cryptographic components, encompassing registered attribute-based encryption and lockable obfuscation.
- *Blind IB-ME (BIB-ME) with certified identities* [18]: Our proposed scheme diverges from BIB-ME with respect to syntax and security definitions. BIB-ME incorporates an extra entity, namely the Identity Certificate Authority, to authenticate users and implement an anonymous key-issuing protocol. This addition renders the syntax and security definition of BIB-ME comparatively more intricate, setting it apart from our proposed scheme.

B. Technical Overview

The main reason for the key escrow problem is that the user has no private input, and the authority knows the secret keys for any identities. To tackle the issue, similar to [5], we assign the dummy attribute sets to the sender and receiver, which are the private inputs and control the encryption and decryption ability of the sender and receiver. Using the specified identity and dummy attribute set, the sender or receiver engages in a fully-simulatable k -out-of n OT protocol with the authority to obtain the encryption key or decryption key. If an encipher box is found to have been maliciously disseminated, it can prove that it is computationally infeasible for the sender to generate it. The only entity that can generate such an encipher box is the authority

as a dishonest one. To achieve this, we share the master key β through a random $d - 1$ degree polynomial when generating the encryption key. Then, by using a k -out-of n OT protocol, the sender can obtain k of the encryption key components, which is oblivious to the authority. The authority cannot masquerade as the sender to generate an encipher box since it does not know which k components the sender has. A similar case happens for abused decoder boxes and receivers as well.

To trace malicious behaviors, we need to check the format of the encryption key and decryption key first. Since β is shared through a $d - 1$ degree polynomial when generating the encryption key, we test its format by randomly selecting a d element set. Because of the splitting technique, the good format of the decryption key can not be simply obtained from the decryption key and public parameter. Here, we test it by constructing a set of components similar to the ciphertext components. When tracing the creator of the encipher box and decoder box, we replace the encryption algorithm and decryption algorithm in the construction with the encipher box and decoder box and then invoke them polynomial times in the security parameter. The generator of the encipher box and decoder box is determined by observing the difference between the decryption or encryption capabilities of the decoder or encipher box and that of the decryption or encryption key.

While we can realize the accountability functionality for malicious behaviours, we also need to realize the message authenticity, bilateral access control and identity privacy. To realize the ciphertext authenticity, we let the sender employ the encryption key to authenticate the ciphertext, and then bind the ciphertext components. Only when the receiver's input is the actual sender's identity in the decryption phase, the correct secret value can be calculated, which realizes bilateral access control and ensures that the identity of the encryption key is identical to that of the ciphertext generator. Whether the secret value is correct or not cannot be determined before decrypting a ciphertext correctly. Otherwise, the sender's identity will be revealed. To achieve this, we embed the secret value into the ciphertext component C , and the receiver can only calculate it under a d element subset I^* . Only after the correct decryption can the receiver confirm the ciphertext authenticity, which also ensures the ciphertext authenticity should not reveal the sender's identity in advance. In [5], the receiver's identity is tested by

$$e(g^{q(i)}, H(\mathbf{u}, ID_B)T_i) = e([H(\mathbf{u}, ID_B)T_i]^{q(i)}, g),$$

where $(g^{q(i)}, [H(\mathbf{u}, ID_B)T_i]^{q(i)})$ are ciphertext components and $(H(\mathbf{u}, ID_B)T_i, g)$ are from the public parameter. To prevent the adversary from getting the receiver's identity through simple tests from the public parameter and ciphertext alone, we employ the splitting technique [19]. Consequently, the decryption key and the corresponding ciphertext components are changed accordingly.

The paper is organized as follows: We first review the related works in Section II. Then, preliminaries are introduced in Section III. After defining A-IB-ME in Section IV, we present the concrete construction in Section V. Next, we prove the security of A-IB-ME in Section VI and evaluate its performance in Section VII. In Section VIII, we present a typical application of A-IB-ME. Finally, we conclude the work in Section IX.

II. RELATED WORKS

In a cryptosystem, the accountability of key abuse and malicious behavior has always been an important problem, such as in the context of attribute-based encryption (ABE) [20], proxy re-encryption [21] and searchable encryption [22]. A-IB-ME mainly considers the problem in IB-ME.

Matchmaking Encryption. ME proposed by Ateniese et al. [1] not only realizes the bilateral access control over encrypted data but also ensures the confidentiality and authenticity of messages as well as the identity privacy of the sender and receiver. In their work [1], the authors presented an efficient IB-ME in a random oracle model. Subsequently, researchers enriched and improved the concept and definition of IB-ME. From a security perspective, Francati et al. [2] and Chen et al. [3] built the IB-ME schemes without relying on random oracles. Boyen et al. [23] also came up with a generic construction of IB-ME with privacy-enhanced features. To counter the threats posed by quantum attacks, post-quantum secure IB-ME schemes [24], [25] have also been proposed. From a functional perspective, Cao et al. [26] built a deniable IB-ME scheme that highlights protecting user privacy in coercion situations. Wu et al. [15] and Sun et al. [16] have expanded the application scope of IB-ME to multiple-receiver scenarios by introducing Fuzzy IB-ME and identity-based broadcast ME. Additionally, Wu et al. [27] tackled the problem of revoking user decryption permissions in multi-receiver scenarios and put forward a relevant IB-ME scheme. From an efficiency standpoint, Huang et al. [28] improved the Fuzzy IB-ME scheme, boosting its efficiency and finishing the security proof in a generic group model. Yang et al. [29] proposed a lightweight IB-ME scheme without bilinear pairs. Due to ME's strict security and privacy restrictions, some researchers relaxed attribute privacy and then realized efficient and flexible bilateral access control [30], [31], [32], [33], [34]. Although many ME schemes and related bilateral access control mechanisms have been proposed, none of the above-mentioned ones can solve the key escrow problem. To address this, Chen et al. [9] implemented an IB-ME scheme in a random oracle model based on the certificateless cryptographic mechanism, but it needs to consider public key substitution attacks. Francati et al. [17] proposed registered ME with a registered attribute encryption mechanism and lockable obfuscation, which also

solves key escrow but can't guarantee message authenticity. Chiku et al. [18] solved the key escrow problem by introducing an Identity Certificate Authority, but also made the solution more complex. Besides, while these solutions [9], [17], [18] solve key escrow, when keys in encipher boxes and decoder boxes are abused, the malicious behaviors can't be held accountable.

Accountable Authority Identity-based Encryption. To mitigate the problem of trust in the authority, Goyal [8] presented the notion of A-IBE and gave an important open problem (i.e., the construction of a opaque A-IBE). Subsequently, Goyal et al. [12] resolved the open question by proposing an A-IBE scheme in a full opaque model, which had a large ciphertext size. To improve the efficiency, Libert et al. [35] designed an efficient A-IBE construction with short ciphertexts and private keys in the selective security model. To overcome the limitation of selective security, Sahai et al. [5] came up with the first A-IBE scheme with full opaque traceability and adaptive security. To avoid inputting the user's decryption key during the tracing process, Lai et al. [14] gave the first A-IBE with public traceability by the public tracing key. Further, Kiayias et al. [13] proposed a generic A-IBE scheme using oblivious transfer, which could be modified to support public traceability. Since IB-ME is more complex than IBE, more concerns should be considered when implementing holding accountable for the key abuse and malicious behavior.

To sum up, based on our analysis of identity-based cryptographic research, as far as we know, no solution can concurrently achieve bilateral access control, ensure message confidentiality and authenticity, protect identity privacy, solve key escrow problem, and hold accountable for the key abuse and malicious behavior. Our proposed A-IB-ME, however, is capable of achieving all these features simultaneously. The introduction of A-IB-ME not only enriches and refines the theoretical framework of IB-ME but also opens up a broader spectrum of possibilities for its practical application and expansion.

III. PRELIMINARIES

A. Notations

Throughout the paper, λ defines the security parameter. For $a \geq b \geq 1$, $[b, a]$ denotes the set of integers $\{b, b+1, \dots, a-1, a\}$. $\mathcal{O}$ expresses the random oracle. $\odot$ records the input of $\mathcal{O}$. If a random algorithm can be terminated within the polynomial step in the security parameter λ for any input, it is a probabilistic polynomial time (PPT) algorithm. If a function vanishes faster than the reciprocal of any polynomial in the security parameter λ , the function is negligible. $ID[j]$ is the j -th bit of the string ID. The lagrange coefficient $\Delta_{i,S}(x)$ for a set S is $\Delta_{i,S}(x) = \prod_{j \in S, j \neq i} \frac{x-j}{i-j}$. The parameter n in the scheme expresses the size of the user identity. k is the size of the user's encryption key or decryption key. d is a threshold value.

B. Bilinear Map

Given two multiplicative groups $\mathbb{G}$ and $\mathbb{G}_T$ of prime order p , where $\mathbb{G}$'s generator is g , an admissible bilinear map $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ has the following properties: for all $u, v \in \mathbb{G}$ and

$a, b \in \mathbb{Z}_p^*$, we have $e(u^a, v^b) = e(u, v)^{ab}$ and non-degeneracy $e(g, g) \neq 1_{\mathbb{G}_T}$, where $1_{\mathbb{G}_T}$ is the identity element.

C. Fully-Simulatable Oblivious Transfer (OT)

A k -out-of- n protocol is a two-party protocol between the sender and receiver, where the sender holds n strings, and at the end, the receiver receives k of them. The receiver gains nothing except k strings and the sender does not know which k strings the receiver acquires. We require that the oblivious transfer protocol is fully-simulatable and meets the Canetti's requirements for universal composability [36]. For more formal definitions and statements, refer to references [37], [38].

D. Decisional Bilinear Diffie-Hellman (DBDH) Assumption

Suppose that $a, b, c, z \in \mathbb{Z}_p^*$ are randomly selected. The DBDH assumption states that no PPT adversary can distinguish between the tuple $(g, A = g^a, B = g^b, C = g^c, e(g, g)^{abc})$ and the tuple $(g, A = g^a, B = g^b, C = g^c, e(g, g)^z)$ with more than a negligible advantage. The formal definition is as follows:

$$|\Pr[A(g^a, g^b, g^c, \hat{e}(g, g)^{abc}) = 1] - \Pr[A(g^a, g^b, g^c, \hat{e}(g, g)^z) = 1]| < \text{negl}(\lambda).$$

E. Decision Linear (D-Lin) Assumption

Given $u, v, h, u^a, v^b, h^c \in \mathbb{G}$, the D-Lin assumption [39] states that no PPT adversary can decide whether $a + b = c$ with more than a negligible advantage, where $a, b, c \in \mathbb{Z}_p^*$ are randomly selected. The formal definition is as follows:

$$|\Pr[A(u, v, h, u^a, v^b, h^c) = 1] - \Pr[A(u, v, h, u^a, v^b, Z) = 1]| < \text{negl}(\lambda),$$

where Z is randomly selected from $\mathbb{G}$.

F. Computational Bilinear Diffie-Hellman (CBDH) Assumption

Suppose that $a, b, c \in \mathbb{Z}_p^*$ are randomly selected. Given the tuple (g, g^a, g^b, g^c) , the CBDH assumption states that no PPT adversary can compute $e(g, g)^{abc}$ with more than a negligible advantage. The formal definition is as follows:

$$\Pr[A(g^a, g^b, g^c) = \hat{e}(g, g)^{abc}] < \text{negl}(\lambda).$$

IV. DEFINITION OF A-IB-ME

A. Syntax of A-IB-ME

Compared with IB-ME, we add the sanity check and tracing algorithm since A-IB-ME provides accountability functionality. In the definition, we do not consider the format check of the ciphertext. The reasons are as follows: The input to the encipher box is a plaintext message, and its output can be decrypted by the well-formed decryption key, which means the ciphertext is well-formed. The ciphertext inputted into the decoder box is generated by calling the algorithm **Enc**, which is well-formed. Besides, additional tracing algorithm can implicate the creator

of an encipher box or a decoder box. Next, A-IB-ME consists of the following polynomial algorithms or protocols:

- $Setup(1^\lambda) \rightarrow (mpk, msk)$: Once inputting the security parameter 1^λ , the algorithm outputs the master public key mpk and master secret key msk , where mpk is the implicit input of other algorithms.
- $EKGen(msk, ID_A)(\rightarrow)ek_{ID_A}(S)$: The process requires the interaction between the authority and the sender. During the interaction process, the sender's input is his identity ID_A , and the authority's private input is msk . At the end, the sender receives an encryption key. Similar to the reference [5], we also use the symbol $(\rightarrow)$ to highlight that the authority may not learn the concrete messages received by the sender.
- $DKGen(msk, ID_B)(\rightarrow)dk_{ID_B}(S')$: The process requires the interaction between the authority and receiver. During the interaction process, the receiver's input is his identity ID_B , and the authority's private input is msk . At the end, the receiver receives a decryption key. Similar to the reference [5], we also use the symbol $(\rightarrow)$ to highlight that the authority may not learn the concrete messages received by the receiver.
- $Enc(M, ek_{ID_A}(S), S'', ID_B) \rightarrow CT_{ID_A, ID_B, S''}$: Once inputting the message M , a random k element subset $S'' \subseteq [1, n]$ as dummy attributes, the encryption key $ek_{ID_A}(S)$, and the designated receiver ID_B , the sender generates the ciphertext $CT_{ID_A, ID_B, S''}$.
- $Dec(CT_{ID'_A, ID'_B, S, S''}, dk_{ID_B}(S'), ID_A) \rightarrow M \text{ or } \perp$: The decryption algorithm executed by the receiver takes the ciphertext $CT_{ID'_A, ID'_B, S, S''}$, the decryption key $dk_{ID_B}(S')$, and the target sender's identity ID_A as input. Then, the receiver can recover the plaintext M if $ID'_A = ID_A$, $ID'_B = ID_B$, $|S \cap S''| \geq d$ and $|S' \cap S''| \geq d$, where d is the highest degree of a random polynomial used in the encryption process; Otherwise, $\perp$ is outputted.
- The format integrity check is divided into the following two cases:
 - $EKSanity(mpk, ek_{ID_A}(S)) \rightarrow \text{Valid or Not-Valid}$: Given the master public key mpk and an encryption key $ek_{ID_A}(S)$, the algorithm checks if $ek_{ID_A}(S)$ is a well-formed encryption key for the identity ID_A and outputs *Valid* or *Not-Valid*.
 - $DKSanity(mpk, dk_{ID_B}(S')) \rightarrow \text{Valid or Not-Valid}$: Given the master public key mpk and a decryption key $dk_{ID_B}(S')$, the algorithm checks if $dk_{ID_B}(S')$ is a well-formed decryption key for the identity ID_B and outputs *Valid* or *Not-Valid*.
- The tracing algorithm includes the following two cases:
 - $Trace^1(\mathcal{E}_{ID_A}, ek_{ID_A}(S)) \rightarrow \text{Sender or authority}$: This algorithm takes as input an ϵ -useful encipher box $\mathcal{E}_{ID_A}$ and encryption key $ek_{ID_A}(S)$, and then outputs the source of the encipher box (i.e., sender or authority).
 - $Trace^2(\mathcal{D}_{ID_B}, dk_{ID_B}(S')) \rightarrow \text{Receiver or authority}$: This algorithm takes as input an ϵ -useful decoder box $\mathcal{D}_{ID_B}$ and decryption key $dk_{ID_B}(S')$, and then outputs the generator of the decoder box (i.e., receiver or authority).

TABLE II
SECURITY GAMES OF PRIVACY AND AUTHENTICITY

$G_{\Pi, \mathcal{A}}^{Priv}(\lambda)$	$G_{\Pi, \mathcal{A}}^{Auth}(\lambda)$
$(mpk, msk) \leftarrow \mathbf{Setup}(1^\lambda)$ $(M_0, M_1, ID_{B_0}^*, ID_{B_1}^*, ID_{A_0}^*, ID_{A_1}^*, st) \leftarrow \mathcal{A}_1^{\mathcal{O}_1, \mathcal{O}_2}(mpk)$, where $ID_{B_0}^*, ID_{B_1}^* \in \mathbb{O}_2$ $b \leftarrow \{0, 1\}$ $ek_{ID_{A_b}^*}(S) \leftarrow \mathbf{EKGen}(msk, ID_{A_b}^*)$ $CT_{ID_{A_b}^*, ID_{B_b}^*, S, S''} \leftarrow \mathbf{Enc}(M_b, ek_{ID_{A_b}^*}(S), ID_{B_b}^*)$ $b' \leftarrow \mathcal{A}_2^{\mathcal{O}_1, \mathcal{O}_2}(mpk, CT_{ID_{A_b}^*, ID_{B_b}^*, S, S''}, st)$ If $(b = b')$ return 1 Else return 0	$(mpk, msk) \leftarrow \mathbf{Setup}(1^\lambda)$ $(CT_{ID_A, ID_B, S, S''}, ID_B^*, ID_A^*) \leftarrow \mathcal{A}^{\mathcal{O}_1, \mathcal{O}_2}(mpk)$ $dk_{ID_B^*}(S') \leftarrow \mathbf{DKGen}(msk, ID_B^*)$ $M \leftarrow \mathbf{Dec}(CT_{ID_A, ID_B, S, S''}, dk_{ID_B^*}(S'), ID_A^*)$ If $ID_A = ID_A^*, ID_B = ID_B^* \forall ID_A \in \mathbb{O}_1$ $ek_{ID_A}(S)$ <i>such that</i> $ S \cap S'' < d$ $\wedge (M \neq \perp)$ return 1 Else return 0

Correctness: The correctness requires that for $\lambda \in \mathbb{N}$, $(mpk, msk) \leftarrow \mathbf{Setup}(1^\lambda)$, $ek_{ID_A}(S) \leftarrow \mathbf{EKGen}(msk, ID_A)$, $dk_{ID_B}(S') \rightarrow \mathbf{DKGen}(msk, ID_B)$,

$$Pr[\mathbf{Dec}(\mathbf{Enc}(M, ek_{ID_A}(S), S'', ID_B), dk_{ID_B}(S'), ID_A) = M] \geq 1 - neg(\lambda),$$

where $ID_A' = ID_A$, $ID_B' = ID_B$, $|S \cap S''| \geq d$, $|S' \cap S''| \geq d$, d is the threshold and S'' is a random set selected in the encryption process. In the above scheme, n is the bit length of the user's identity. k is the length of the encryption and decryption keys. Note that by setting the appropriate parameters k and d , we can ensure that given some set $S \subseteq [1, n]$ such that $|S| = k$, for a random k element set $S' \subseteq [1, n]$, we have

$$Pr[|S \cap S'| \geq d] > 1 - \mu(\lambda),$$

where $\mu(\lambda)$ is some negligible function. From this point, assume k and d are initialized thusly. For detailed parameter settings, please refer to [5].

ϵ -Useful Encipher Box: For a non-negligible ϵ , a PPT algorithm $\mathcal{E}_{ID_A}$ is an ϵ -useful encipher box for the identity ID_A if

$$Pr[M \leftarrow \mathcal{M} : \mathbf{Dec}(\mathcal{E}_{ID_A}(M, S'', ID_B), dk_{ID_B}(S'), ID_A) = M] \geq \epsilon,$$

where ID_B is a random receiver and $|S'' \cap S'| > d$.

ϵ -Useful Decoder Box: For a non-negligible ϵ , a PPT algorithm $\mathcal{D}_{ID_B}$ is an ϵ -useful decoder box for the identity ID_B if

$$Pr[M \leftarrow \mathcal{M} : \mathcal{D}_{ID_B}(\mathbf{Enc}(M, ek_{ID_A}(S), S'', ID_B), S', ID_A) = M] \geq \epsilon,$$

where ID_A is a random sender, $|S'' \cap S| > d$ and $|S'' \cap S'| > d$.

B. Security of A-IB-ME

While A-IB-ME realizes the functionality of IB-ME, it can also be accountable for the key abuse. Accordingly, A-IB-ME not only guarantees the privacy and authenticity, but also ensures the security of dishonest senders, dishonest receivers and dishonest authority.

Privacy of A-IB-ME: Privacy captures that no information is leaked to non-intended receivers, including the message, the identities of the sender and receiver, whose definition is presented in Table II. Besides, we do not consider the privacy of

the users' dummy attributes since they have nothing to do with identity information. $\mathcal{O}_1$ and $\mathcal{O}_2$ are the encryption oracle and decryption oracle respectively. $\mathbb{O}_1$ and $\mathbb{O}_2$ records the access to $\mathcal{O}_1$ and $\mathcal{O}_2$. st denotes the state information $\mathcal{A}$ gets after finishing the first query phase. In $G_{\Pi, \mathcal{A}}^{Priv}(\lambda)$, $(ID_{B_0}^*, ID_{A_0}^*)$ and $(ID_{B_1}^*, ID_{A_1}^*)$ in the challenge message submitted by $\mathcal{A}$ should have once queried, which simulates that $\mathcal{A}$ attacks some ciphertext associated with some actual receiver (i.e., $ID_{B_1}^*$ or $ID_{B_0}^*$). S and S'' are two k element random set, which is used in the encryption key generation and encryption process respectively.

Definition 1 (Privacy of A-IB-ME): We say that A-IB-ME achieves privacy if for any valid PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ we have:

$$|Pr[G_{\Pi, \mathcal{A}}^{Priv}(\lambda) = 1] - \frac{1}{2}| \leq negl(\lambda),$$

where $\mathcal{A}$ is valid if $dk_{ID_{B_0}^*}(S'_0)$ and $dk_{ID_{B_1}^*}(S'_1)$ such that $|S'' \cap S'_0| < d$ and $|S'' \cap S'_1| < d$.

Authenticity of A-IB-ME: Authenticity is resilient to malicious senders, which is shown in Table II. In $G_{\Pi, \mathcal{A}}^{Auth}(\lambda)$, the adversary $\mathcal{A}$ can query all encryption keys for $ID_A \neq ID_A^*$. If $ID_A = ID_A^*$, $|S \cap S''|$ must be less than d . A forged ciphertext $(CT_{ID_A, ID_B, S, S''}, ID_B^*, ID_A^*)$ is valid if it can be decrypted by

$$\mathbf{Dec}(CT_{ID_A, ID_B, S, S''}, dk_{ID_B^*}(S'), ID_A^*) \neq \perp,$$

where $ID_A = ID_A^*$, $ID_B = ID_B^*$, $ID_A \in \mathbb{O}_1$ and $ek_{ID_A}(S)$ such that $|S \cap S''| < d$. In $G_{\Pi, \mathcal{A}}^{Auth}(\lambda)$, the encryption key on ID_A should have been queried by $\mathcal{A}$ once, which simulates that $\mathcal{A}$ forges a ciphertext associated with some actual sender (i.e., ID_A).

Definition 2 (Authenticity of A-IB-ME): We say that A-IB-ME satisfies authenticity if for any PPT adversary $\mathcal{A}$ we have:

$$Pr[G_{\Pi, \mathcal{A}}^{Auth}(\lambda) = 1] \leq negl(\lambda).$$

Dishonest-Sender Security of A-IB-ME: The dishonest sender security captures the setting where a dishonest sender constructs an encipher box $\mathcal{E}_{ID_A}$, but the tracing algorithm implicates the authority. The formal definition is presented in Table III. The adversary $\mathcal{A}$ can query the encryption key or decryption key for ID_i , and then output an encryption key $ek_{ID_A}(S)$ that has been issued and its corresponding encipher box $\mathcal{E}_{ID_A}$. After inputting $ek_{ID_A}(S)$ and $\mathcal{E}_{ID_A}$, the game outputs 1 if the tracing algorithm implicates the authority; Otherwise the game outputs 0.

TABLE III
SECURITY GAMES OF DISHONEST-SENDER/RECEIVER SECURITY

$G_{\Pi, \mathcal{A}}^{DisSender}(\lambda)$	$G_{\Pi, \mathcal{A}}^{DisRev}(\lambda)$
$(mpk, msk) \leftarrow \mathbf{Setup}(1^\lambda)$ $(\mathcal{E}_{ID_A}, ek_{ID_A}(S)) \leftarrow \mathcal{A}^{\mathcal{O}_1, \mathcal{O}_2}(mpk)$ $Trace^1(\mathcal{E}_{ID_A}, ek_{ID_A}(S)) = 1$ or 0	$(mpk, msk) \leftarrow \mathbf{Setup}(1^\lambda)$ $(\mathcal{D}_{ID_B}, dk_{ID_B}(S')) \leftarrow \mathcal{A}^{\mathcal{O}_1, \mathcal{O}_2}(mpk)$ $Trace^2(\mathcal{D}_{ID_B}, dk_{ID_B}(S')) = 1$ or 0

TABLE IV
SECURITY GAMES OF DISHONEST-AUTHORITY SECURITY

$G_{\Pi, \mathcal{A}}^{DisAuth1}(\lambda)$	$G_{\Pi, \mathcal{A}}^{DisAuth2}(\lambda)$
$(mpk, ID_A) \leftarrow \mathcal{A}(1^\lambda)$ $ek_{ID_A}(S) \leftarrow \mathcal{S}^{\mathcal{O}_1(ID_A, \cdot), \mathcal{O}_2}(mpk, S)$ $\mathcal{E}_{ID_A} \leftarrow \mathcal{A}^{\mathcal{O}_3(ek_{ID_A}(S), \cdot)}(mpk)$ $Trace^1(\mathcal{E}_{ID_A}, ek_{ID_A}(S)) = 1$ or 0	$(mpk, ID_B) \leftarrow \mathcal{A}(1^\lambda)$ $dk_{ID_B}(S') \leftarrow \mathcal{S}^{\mathcal{O}_1, \mathcal{O}_2(ID_B, \cdot)}(S')$ $\mathcal{D}_{ID_B} \leftarrow \mathcal{A}^{\mathcal{O}_4(dk_{ID_B}(S'), \cdot)}(mpk)$ $Trace^2(\mathcal{D}_{ID_B}, dk_{ID_B}(S')) = 1$ or 0

Definition 3 (Dishonest-Sender Security of A-IB-ME): We say that A-IB-ME satisfies the dishonest-sender security if for any PPT adversary $\mathcal{A}$ we have:

$$Pr[G_{\Pi, \mathcal{A}}^{DisSender}(\lambda) = 1] \leq \text{negl}(\lambda).$$

Dishonest-Receiver Security of A-IB-ME: The dishonest-receiver security captures the setting where a dishonest receiver constructs a decoder box $\mathcal{D}_{ID_B}$, but the tracing algorithm implicates the authority. The formal definition is presented in Table III. The adversary $\mathcal{A}$ can query the encryption key or decryption key for ID_i , and then output a decryption key $dk_{ID_B}(S)$ that has been issued and its corresponding decoder box $\mathcal{D}_{ID_B}$. After inputting $dk_{ID_B}(S)$ and $\mathcal{D}_{ID_B}$, the game outputs 1 if the tracing algorithm implicates the authority; Otherwise the game outputs 0.

Definition 4 (Dishonest-Receiver Security of A-IB-ME): We say that A-IB-ME satisfies the dishonest-receiver security if for any PPT adversary $\mathcal{A}$ we have:

$$Pr[G_{\Pi, \mathcal{A}}^{DisRev}(\lambda) = 1] \leq \text{negl}(\lambda).$$

Dishonest-Authority Security of A-IB-ME: The dishonest-authority security captures the setting where a dishonest authority constructs an encipher box or a decoder box, but the tracing algorithm implicates the sender or receiver. There are two behaviors of dishonest authority. The formal definition of the dishonest-authority security $G_{\Pi, \mathcal{A}}^{DisAuth}(\lambda) = (G_{\Pi, \mathcal{A}}^{DisAuth1}(\lambda), G_{\Pi, \mathcal{A}}^{DisAuth2}(\lambda))$ is presented in Table IV. $\mathcal{O}_3$ and $\mathcal{O}_4$ denote the encryption oracle and decryption oracle, respectively. $\mathcal{S}^{\mathcal{O}_1(ID_A, \cdot)}$ means that the simulator $\mathcal{S}$ queries the encryption key on the identity ID_A . $\mathcal{A}^{\mathcal{O}_3(ek_{ID_A}(S), \cdot)}$ means $\mathcal{A}$ queries the ciphertext on the identity ID_A . $\mathcal{S}^{\mathcal{O}_2(ID_B, \cdot)}$ means that the simulator $\mathcal{S}$ queries the decryption key on the identity ID_B . $\mathcal{A}^{\mathcal{O}_4(dk_{ID_B}(S'), \cdot)}$ means $\mathcal{A}$ queries the decryption of the ciphertext associated with the identity ID_B . In the final phase, the game outputs 1 if the tracing algorithm implicates the sender or receiver; Otherwise the game outputs 0.

Definition 5 (Dishonest-Authority Security of A-IB-ME): We say that A-IB-ME satisfies the dishonest-authority security if for any PPT adversary $\mathcal{A}$ we have:

$$Pr[G_{\Pi, \mathcal{A}}^{Dis-Auth}(\lambda) = 1] \leq \text{negl}(\lambda).$$

Definition 6 (Secure A-IB-ME): We say that A-IB-ME Π is semantically secure if Π meets the privacy (Definition 1), authenticity (Definition 2), dishonest-sender security (Definition 3), dishonest-receiver security (Definition 4) and dishonest-authority security (Definition 5).

V. CONSTRUCTION OF A-IB-ME

A. Concrete Construction

- Setup:** After inputting the security parameter 1^λ , the authority selects a bilinear map $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$, random numbers $\alpha, \beta, t_1, t_2, t_3, t_4 \leftarrow \mathbb{Z}_p^*$, two random elements $g_2, g_3 \leftarrow \mathbb{G}$, four random vectors $\mathbf{T}[i], \mathbf{T}'[i] \leftarrow \mathbb{G}, \mathbf{u}[i], \mathbf{u}'[i] \leftarrow \mathbb{G}$ for $i \in [0, n]$, a cryptographic hash function $H_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_p^*$ and computes

$$g_1 = g^\alpha, g'_1 = g^\beta, Y_1 = e(g_1, g_2)^{t_1 t_2}, Y_2 = e(g_3, g)^\beta,$$

$$v_1 = g^{t_1}, v_2 = g^{t_2}, v_3 = g^{t_3}, v_4 = g^{t_4}.$$

Given an identity $ID \in \{0, 1\}^n$ and $\mathbf{u}[i], \mathbf{u}'[i] \in \mathbb{G}$ for $i \in [0, n]$, the authority sets two functions

$$H(\mathbf{u}, ID) = \mathbf{u}[0] \Pi_{j \in [1, n]} \mathbf{u}[j]^{ID[j]},$$

and

$$H'(\mathbf{u}', ID) = \mathbf{u}'[0] \Pi_{j \in [1, n]} \mathbf{u}'[j]^{ID[j]}.$$

Finally, the master public key and master secret key are mpk and msk , where $mpk = (g_1, g'_1, g_2, g_3, Y_1, Y_2, v_1, v_2, v_3, v_4, \mathbf{u}, \mathbf{T}, \mathbf{u}', \mathbf{T}', H_1)$ and $msk = (g_2^\alpha, \beta, t_1, t_2, t_3, t_4)$. Without causing confusion, we write $\mathbf{T}[i]$ as T_i and $\mathbf{T}'[i]$ as T'_i .

- EKGen:** This is an interactive algorithm between the authority and sender. After inputting the master secret key msk and the sender's identity ID_A , the authority randomly selects $r_i \in \mathbb{Z}_p^*$, a random $d-1$ degree polynomial $j(x)$ such that $j(0) = \beta$ and generates the universal encryption key

$$\begin{aligned} ek_{ID_A} &= (ek_{ID_A, i1}, ek_{ID_A, i2}) \\ &= (g_3^{j(i)} [H'(\mathbf{u}', ID_A) T'_i]^{r_i}, g^{r_i}), \end{aligned}$$

for $i \in [1, n]$. Next, it randomly permutes them by a permutation function $\sigma : [n] \rightarrow [n]$. Then, the authority

engages in a fully-simulatable k -out-of- n OT protocol with the sender, where the private input to the authority is the permuted set on ek_{ID_A} and the private input to the sender is a random k -element subset $S \subset [1, n]$, which is independent of identity information. At the end, the sender receives the encryption key $ek'_{ID_A}(S) = \{g_3^{j(i)}[H'(\mathbf{u}', ID_A)T_i']^{r_i}, g^{r_i}\}_{i \in \sigma(S)}$. Besides, the authority also sends σ to the sender in the clear. Finally, the sender obtains the encryption key in original order

$$\begin{aligned} ek_{ID_A}(S) &= \{ek_{ID_A,i1}, ek_{ID_A,i2}\}_{i \in S} \\ &= \{g_3^{j(i)}[H'(\mathbf{u}', ID_A)T_i']^{r_i}, g^{r_i}\}_{i \in S}. \end{aligned}$$

- **DKGen:** This is an interactive algorithm between the authority and receiver. After inputting the master secret key msk and the receiver's identity ID_B , the authority randomly selects $k_{i1}, k_{i2} \in \mathbb{Z}_p^*$ for $i \in [1, n]$ and computes

$$dk_{ID_B} = \begin{cases} dk_{ID_B,i1} = g^{k_{i1}t_1t_2+k_{i2}t_3t_4} \\ dk_{ID_B,i2} = g_2^{-\alpha t_2}[H(\mathbf{u}, ID_B)T_i]^{-k_{i1}t_2} \\ dk_{ID_B,i3} = g_2^{\alpha t_1}[H(\mathbf{u}, ID_B)T_i]^{k_{i1}t_1} \\ dk_{ID_B,i4} = [H(\mathbf{u}, ID_B)T_i]^{k_{i2}t_4} \\ dk_{ID_B,i5} = [H(\mathbf{u}, ID_B)T_i]^{k_{i2}t_3} \end{cases}$$

Next, it randomly permutes them by a permutation function $\sigma' : [n] \rightarrow [n]$. Then, the authority engages in a fully-simulatable k -out-of- n OT protocol with the receiver, where the privacy inputs to the authority and the receiver are the permuted set on dk_{ID_B} and a random k -element subset $S' \subset [1, n]$, respectively, where S' is independent of identity information. At the end, the receiver receives the decryption key $dk'_{ID_B}(S') = \{dk_{ID_B,i1}, dk_{ID_B,i2}, dk_{ID_B,i3}, dk_{ID_B,i4}, dk_{ID_B,i5}\}_{i \in \sigma'(S')}$. Besides, the authority also sends σ' to the receiver in the clear. Finally, the receiver obtains the decryption key in original order $dk_{ID_B}(S') = \{dk_{ID_B,i1}, dk_{ID_B,i2}, dk_{ID_B,i3}, dk_{ID_B,i4}, dk_{ID_B,i5}\}_{i \in S'}$.

- **Enc:** After inputting the message $M \in \mathbb{G}_T$, the encryption key $ek_{ID_A}(S)$, and the designated receiver ID_B , the sender selects a random k element set $S'' \subseteq [1, n]$, random numbers $s, s_{i1}, s_{i2} \in \mathbb{Z}_p^*$ and a $(d-1)$ -degree polynomials $q(x)$ such that $q(0) = s$. Then, the sender computes

$$\begin{aligned} K_s &= Y_1^s, K_l = Y_2^s, C = M \cdot K_s \cdot K_l, C_{i1} \\ &= [H(\mathbf{u}, ID_B)T_i]^{q(i)}, \end{aligned}$$

$$C_{i2} = v_1^{q(i)-s_{i1}}, C_{i3} = v_2^{s_{i1}}, C_{i4} = v_3^{q(i)-s_{i2}}, C_{i5} = v_4^{s_{i2}}$$

for $i \in S''$. Next, for $i \in S$, the sender also selects randomly $z_i, z'_i \in \mathbb{Z}_p^*$ and computes

$$C_{i6} = g^{z'_i}, C_{i7} = (ek_{ID_A,i2}(S) \cdot g^{z_i})^s,$$

and $C_{i8} = (ek_{ID_A,i1}(S))^s \cdot [H'(\mathbf{u}', ID_A)T_i']^{s \cdot z_i} \cdot H_1(C_{i1}||C_{i2}||C_{i3}||C_{i4}||C_{i5}||C_{i6}||C_{i7})^{z'_i}$. Then, the sender calculates the intersection $\mathbb{I} = S \cap S''$. If $|\mathbb{I}| \geq d$, the sender randomly selects a d element subset $I^* \subseteq \mathbb{I}$. If $|\mathbb{I}| < d$, the sender reselects the dummy attribute set S'' and

recalculates the ciphertext. Finally, the ciphertext is $CT = (S'', I^*, C, \{C_{i1}, C_{i2}, C_{i3}, C_{i4}, C_{i5}, C_{i6}, C_{i7}, C_{i8}\}_{i \in [1, k]})$.

- **Dec:** After inputting the expected sender's identity ID_A , the decryption key $dk_{ID_B}(S')$ associated with the receiver ID_B , and the ciphertext CT associated with ID'_A and ID'_B , the receiver first denotes $CT_i = C||C_{i1}||C_{i2}||C_{i3}||C_{i4}||C_{i5}||C_{i6}||C_{i7}$, $HT'_i = [H'(\mathbf{u}', ID_A)T_i']$ and computes

$$\begin{aligned} K'_l &= \prod_{i \in I^*} \left(\frac{e(C_{i8}, g)}{e([HT'_i], C_{i7})e(H_1(CT_i), C_{i6})} \right)^{\Delta_{i,I^*}(0)} \\ &= \prod_{i \in I^*} \left(\frac{e((g_3^{j(i)}[H'(\mathbf{u}', ID_A)T_i']^{r_i})^s [HT'_i]^{s \cdot z_i} \cdot H_1(CT_i)^{z'_i}, g)}{e([HT'_i], (g^{r_i} \cdot g^{z_i})^s)e(H_1(CT_i), g^{z'_i})} \right)^{\Delta_{i,I^*}(0)} \\ &= \prod_{i \in I^*} e(g_3, g^s)^{j(i)\Delta_{i,I^*}(0)} = e(g_3, g)^{s\beta}. \end{aligned}$$

Then, the receiver takes a d element subset $I \subseteq (S' \cap S'')$, denotes $HT_i = H(\mathbf{u}, ID_B)T_i$ and computes

$$\begin{aligned} K'_s &= \prod_{i \in I} \left(\frac{e(C_{i1}, dk_{ID_B,i1})e(C_{i2}, dk_{ID_B,i2})}{e(C_{i3}, dk_{ID_B,i3})e(C_{i4}, dk_{ID_B,i4})e(C_{i5}, dk_{ID_B,i5})} \right)^{\Delta_{i,I}(0)} \\ &= \prod_{i \in I} \left(\frac{e([HT_i]^{q(i)}, g^{k_{i1}t_1t_2+k_{i2}t_3t_4})}{e(v_2^{s_{i1}}, g_2^{\alpha t_1}[HT_i]^{k_{i1}t_1})} \right. \\ &\quad \times \left. \frac{e(v_1^{q(i)-s_{i1}}, g_2^{-\alpha t_2}[HT_i]^{-k_{i1}t_2})}{e(v_3^{q(i)-s_{i2}}, [HT_i]^{k_{i2}t_4})e(v_4^{s_{i2}}, [HT_i]^{k_{i2}t_3})} \right)^{\Delta_{i,I}(0)} \\ &= \prod_{i \in I} \left(\frac{e([HT_i]^{q(i)}, g^{k_{i1}t_1t_2+k_{i2}t_3t_4})}{e(g^{t_2s_{i1}}, g_2^{\alpha t_1}[HT_i]^{k_{i1}t_1})} \right. \\ &\quad \times \left. \frac{e(g^{t_1(q(i)-s_{i1})}, g_2^{-\alpha t_2}[HT_i]^{-k_{i1}t_2})}{e(g^{t_3(q(i)-s_{i2})}, [HT_i]^{k_{i2}t_4})e(g^{t_4s_{i2}}, [HT_i]^{k_{i2}t_3})} \right)^{\Delta_{i,I}(0)} \\ &= \prod_{i \in I} (e(g, g_2^\alpha)^{-t_1t_2q(i)})^{\Delta_{i,I}(0)} = e(g_1, g_2)^{-t_1t_2s}. \end{aligned}$$

The receiver computes $M = C \cdot K'_s / K'_l$ if $ID'_A = ID_A$, $ID'_B = ID_B$, $|S \cap S''| \geq d$ and $|S' \cap S''| \geq d$.

- The format integrity check is divided into the following two cases:

- **EKeySanity:** For the encryption key components $(g_3^{j(i)}[H'(\mathbf{u}', ID_A)T_i']^{r_i}, g^{r_i})_{i \in S}$, if

$$\prod_{i \in \mathbb{S}} \left(\frac{e(g_3^{j(i)}[H'(\mathbf{u}', ID_A)T_i']^{r_i}, g)}{e([H'(\mathbf{u}', ID_A)T_i']^{r_i}, g^{r_i})} \right)^{\Delta_{i,\mathbb{S}}(0)} = e(g_3, g'_1)$$

for any d element subset $\mathbb{S} \subseteq S$, the encryption key is well-formed.

- **DKeySanity:** For the decryption key components,

$$dk_{ID_B}(S') = \begin{cases} dk_{ID_B,i1} = g^{k_{i1}t_1t_2+k_{i2}t_3t_4} \\ dk_{ID_B,i2} = g_2^{-\alpha t_2}[H(\mathbf{u}, ID_B)T_i]^{-k_{i1}t_2} \\ dk_{ID_B,i3} = g_2^{\alpha t_1}[H(\mathbf{u}, ID_B)T_i]^{k_{i1}t_1} \\ dk_{ID_B,i4} = [H(\mathbf{u}, ID_B)T_i]^{k_{i2}t_4} \\ dk_{ID_B,i5} = [H(\mathbf{u}, ID_B)T_i]^{k_{i2}t_3} \end{cases}$$

the check process is as follows: one selects random numbers $s_{i1}, s_{i2} \in \mathbb{Z}_p^*$, computes $D_{i1} = [H(\mathbf{u}, ID_B)T_i]$, $D_{i2} = v_1^{1-s_{i1}}$, $D_{i3} = v_2^{s_{i1}}$, $D_{i4} = v_3^{1-s_{i2}}$, and $D_{i5} = v_4^{s_{i2}}$ for $i \in S'$, denotes $HT_i = [H(\mathbf{u}, ID_B)T_i]$ and tests the following equation:

$$\begin{aligned} & \frac{e(D_{i1}, dk_{ID_B, i1})e(D_{i2}, dk_{ID_B, i2})}{e(D_{i3}, dk_{ID_B, i3})e(D_{i4}, dk_{ID_B, i4})e(D_{i5}, dk_{ID_B, i5})} \\ &= \frac{e(HT_i, g^{k_{i1}t_1t_2+k_{i2}t_3t_4})}{e(v_2^{s_{i1}}, g_2^{\alpha t_1} [HT_i]^{k_{i1}t_1})e(v_3^{1-s_{i2}}, [HT_i]^{k_{i2}t_4})} \\ &= \frac{e(v_1^{1-s_{i1}}, g_2^{-\alpha t_2} [HT_i]^{-k_{i1}t_2})}{e(v_4^{s_{i2}}, [HT_i]^{k_{i2}t_3})} \\ &= \frac{e(HT_i, g^{k_{i1}t_1t_2+k_{i2}t_3t_4})}{e(g^{t_2s_{i1}}, g_2^{\alpha t_1} [HT_i]^{k_{i1}t_1})e(g^{t_3(1-s_{i2})}, [HT_i]^{k_{i2}t_4})} \\ &= \frac{e(g^{t_1(1-s_{i1})}, g_2^{-\alpha t_2} [HT_i]^{-k_{i1}t_2})}{e(g^{t_4(s_{i2})}, [HT_i]^{k_{i2}t_3})} \\ &= e(g, g_2^\alpha)^{-t_1t_2} = e(g_1, g_2)^{-t_1t_2\alpha} = Y_1^{-1}. \end{aligned}$$

If the above equations hold for $i \in S'$, the decryption key components are well-formed.

- The tracing algorithm includes the following two cases:
- **Trace¹**($\mathcal{E}_{ID_A}, ek_{ID_A}(S)$) $\rightarrow$ Sender or authority: After inputting an encryption key $ek_{ID_A}(S)$ and an encipher box $\mathcal{E}_{ID_A}$, the tracing algorithm runs the algorithm **EKeySanity** to check the validity of the encryption key and repeats the following experiment:
 - Select a random message $M \in \mathbb{G}_T$, a random receiver ID_B and a k element set $S'' \subseteq [1, n]$ such that $|S \cap S''| < d$. Then, $\mathcal{E}_{ID_A}$ outputs some $CT_{ID_A, ID_B, S, S''} = \mathcal{E}_{ID_A}(M, ID_B, S'')$.
 - Select a k element set $S' \subseteq [1, n]$ such that $|S'' \cap S'| \geq d$ and output the decryption key $dk_{ID_B}(S')$.
 - Output $M' = \text{Dec}(CT_{ID_A, ID_B, S, S''}, dk_{ID_B}(S'), ID_A)$
 - If for any iteration $M' = M$, the tracing algorithm outputs the authority; Otherwise, it outputs the sender.
- **Trace²**($\mathcal{D}_{ID_B}, dk_{ID_B}(S')$) $\rightarrow$ Receiver or authority: After inputting a decryption key $dk_{ID_B}(S')$ and a decoder box $\mathcal{D}_{ID_B}$, the tracing algorithm runs the algorithm **DKeySanity** to check the validity of the decryption key and repeats the following experiment:
 - Select a random message $M \in \mathbb{G}_T$, a random sender ID_A , a k element set $S'' \subseteq [1, n]$ such that $|S' \cap S''| < d$.
 - Obtain some $CT_{ID_A, ID_B, S, S''} = \text{Enc}(M, ek_{ID_A}(S), ID_B, S'')$ with $|S'' \cap S| \geq d$, where $ek_{ID_A}(S)$ is generated as the construction does.
 - The decoder box outputs a message of $CT_{ID_A, ID_B, S, S''}$, $M' = \mathcal{D}_{ID_B}(CT_{ID_A, ID_B, S, S''}, S', ID_A)$

If for any iteration $M' = M$, the tracing algorithm outputs the authority; Otherwise, it outputs the recipient.

VI. PROOF OF A-IB-ME

Theorem 1: If the DBDH, D-Lin, CBDH assumptions hold in $(\mathbb{G}, \mathbb{G}_T, e)$, A-IB-ME is semantically secure.

Proof. Due to space limitations, we have included the proof of Theorem 1 in the Appendix, available online.

VII. PERFORMANCE EVALUATION

In this section, we first theoretically analyze the efficiency of A-IB-ME, and then we implement it to more intuitively show its efficiency.

Table V shows the performance comparison among A-IB-ME, IB-ME [3] and A-IBE [5] in terms of the computation costs of each phase and the storage costs of the main components, where A-IB-ME is derived from A-IBE [5]. As shown, compared to A-IBE [5] and A-IB-ME, the efficiency of IB-ME [3] remains constant in terms of computation overheads and storage costs. This is primarily due to the fact that in IB-ME [3] the user's identity is identified by a string whereas in A-IBE [5] and A-IB-ME, it is identified by an attribute set. It can be seen from the table that the computation overheads of encryption key generation and decryption key generation in A-IB-ME are linearly related to the number of dummy attributes n , which is a property inherited from A-IBE where the computation costs of decryption keys are also linearly related to the number of dummy attributes n . Additionally, the efficiency of their generation is affected by the computation cost of a k -out-of- n OT protocol. The computation cost of the encryption process are linearly related to the number of dummy attributes k while the decryption process has linear relation to the degree of polynomial d in A-IB-ME and A-IBE. From the table, we can also observe that the storage costs of the encryption key and decryption key are linearly connected with the number of dummy attributes k in two schemes. The storage cost of ciphertexts in A-IB-ME and A-IBE is correlated with both the dummy attributes k and n , but the effect of the dummy attribute k is greater. From the table, it is apparent that A-IBE outperforms A-IB-ME regarding computation overheads and storage costs. However A-IB-ME provides richer functionalities, such as additional bilateral access control, message authenticity, and identity privacy, which results in increased computational burdens and storage costs. Nonetheless, the performance of both schemes is on the same order of magnitude and additional functionalities provided by A-IB-ME compensates for the lower performance, which represents a reasonable trade-off for achieving stronger security and more comprehensive functionalities. Overall, IB-ME and A-IBE outperform A-IB-ME in terms of efficiency, yet they are inferior to it in terms of functional richness and security enhancement.

To evaluate the performance of A-IB-ME more intuitively and accurately, we implement it¹ on Windows 10 Pro 21H2 with 11th Gen Intel(R) Core(TM) i7-11800H CPU 2.30 GHz 8 cores, 24 GB RAM, 512 GB SSD, and 1024 GB SSD based on the default Type A elliptic curve of JPBC 2.0.0., where we employ the protocol in [40] to instantiate OT. Additionally, the bit length

¹<https://github.com/yueryang/Cryptographic-Schemes/blob/main/SchemeAAIBME/>.

TABLE V
PERFORMANCE COMPARISON BETWEEN A-IB-ME AND A-IBE

Schemes	Computation Costs				Storage Costs		
	EKGen	DKGen	Encryption	Decryption	EKGen	DKGen	Ciphertext
A-IBE [5]	—	$3nT_E + T_{OT}$	$(1 + 2k)T_E$	$2dT_P + dT_E$	—	$2k \mathbb{G} $	$d \mathbb{Z}_p + \mathbb{G}_T + 2k \mathbb{G} + n \text{ bit}$
IB-ME [3]	$8T_e$	$17T_e$	$9T_e$	$8T_P + T_e$	$8 \mathbb{G} $	$17 \mathbb{G} $	$9 \mathbb{G} $
A-IB-ME	$3nT_E + T_{OT}$	$(5n + 2)T_E + T_{OT}$	$(2 + 11k)T_E$	$8dT_P + 2dT_E$	$2k \mathbb{G} $	$5k \mathbb{G} $	$(k + d) \mathbb{Z}_p + \mathbb{G}_T + 8k \mathbb{G} $

T_E , T_P and T_{OT} respectively represent the computation costs of an exponential operation in $\mathbb{G}$ or $\mathbb{G}_T$, a bilinear mapping operation and a k -out-of- n OT operation. $|\mathbb{G}|$, $|\mathbb{G}_T|$ and $|\mathbb{Z}_p|$ represent the size of elements in $\mathbb{G}$, $\mathbb{G}_T$ and $\mathbb{Z}_p$, respectively.

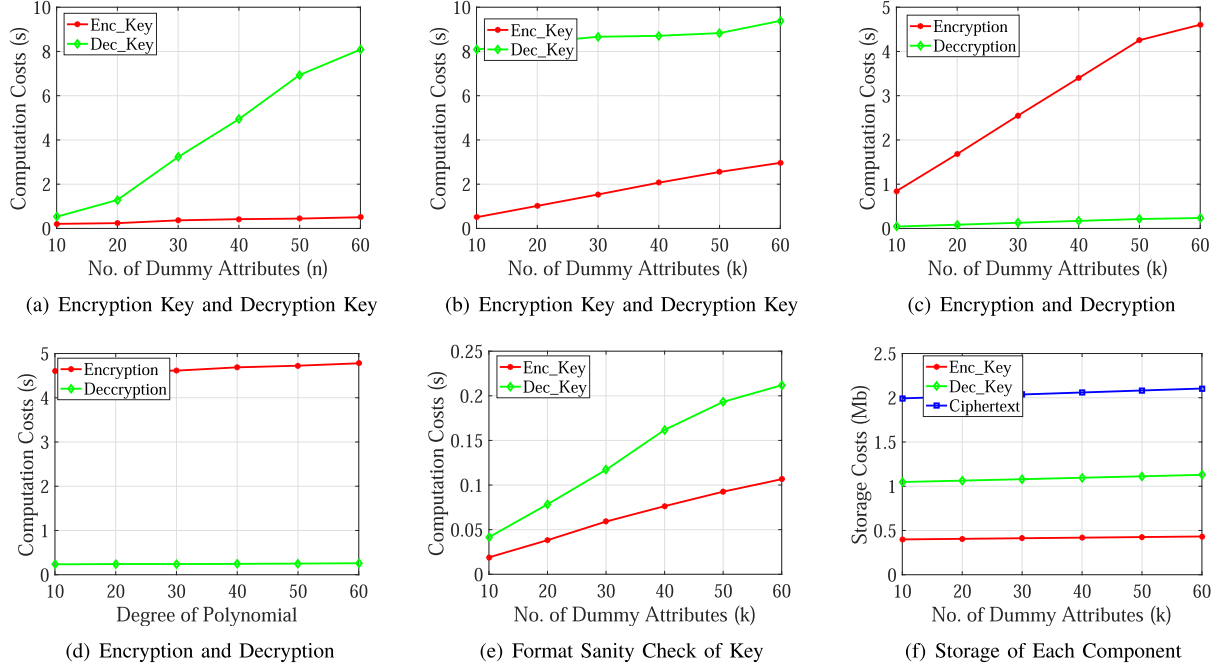


Fig. 1. Performance evaluation.

of p is 160 bits, and the order of $\mathbb{G}$ is 512 bits, which means that the security parameter is 160 bits in elliptic curve cryptography. To eliminate the interference of random factors, we estimate that the computation cost of each phase is an average of 20 repeats. Fig. 1 depicts the performance evaluation of A-IB-ME. Figs. 1(a) and 1(b) show the connection between the key generation process and dummy attributes n and k , where $d = 10$ and $k = 10$ in Fig. 1(a) and $d = 10$ and $n = 60$ in Fig. 1(b). From the figures, the computation costs of key generation rise with the increase of dummy attributes n and k . Besides, the dummy attribute n has a relatively greater effect on the computation cost of the key generation process. In practical applications, we can employ more efficient OT for instantiation. Fig. 1(c) exhibits the relation between the encryption-decryption process and the dummy attribute k , where $d = 10$ and $n = 60$. Fig. 1(d) shows the connection between the encryption-decryption process and the degree of polynomial d , where $k = 60$ and $n = 60$. It can be seen from the figure that the effect of k on the encryption-decryption process is greater than that of d on the encryption-decryption process. The dummy attribute k increases the number of the exponential operation in the encryption process while the dummy attribute d increases the number of the bilinear pairing operation in the decryption process. Fig.

1(e) displays the effect of the dummy attribute k on the format sanity check of encryption-decryption key, where $d = 10$ and $n = 60$. As you can see from the figure, computation costs of key format sanity check are linearly related to the dummy attribute k . The sizes of the encryption key and decryption key affect the computation costs of key format sanity check, which is associated with k . Fig. 1(f) presents the storage costs of different components under different values of the dummy attribute k , where $d = 10$ and $n = 60$. As can be seen from the figure, the storage costs increase linearly with the dummy attribute k . That is mainly because their sizes are mainly related to the dummy attribute k . As for the efficiency of the tracing algorithm, it can be obtained from the efficiency of the format sanity check and encryption-decryption process.

Enhanced functionality and security typically lead to higher computational and communication overheads, which inevitably trade off some efficiency. From the perspective of functional equivalence, an integrated solution rather than a combination of multiple schemes offers users a more convenient and efficient option. For instance, A-IBE has higher computational overheads than IBE but provides richer functionalities and stronger security. To meet the same functional requirements (i.e., bilateral access control, identity privacy, message authenticity, message

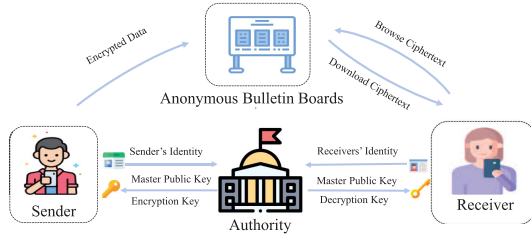


Fig. 2. System model of user matching.

confidentiality and accountability for malicious behaviors), A-IBE needs additional techniques to achieve identity privacy, message authenticity and bilateral access control, while IB-ME requires supplementary techniques for malicious behavior accountability. Currently, there are no clear technical standards for extending A-IBE and IB-ME to realize these functionalities, and their performance after extension cannot be directly evaluated. Though A-IB-ME has higher computational overheads than other related primitives, it uniquely achieves accountability for malicious behaviors unavailable in other schemes. In scenarios requiring the above-mentioned comprehensive functionalities, A-IB-ME serves as a convenient and efficient integrated solution, which is a balanced choice among security, efficiency and functionality.

VIII. APPLICATIONS

A-IB-ME provides bilateral access control, message authenticity and identity privacy of the sender and receiver while achieving accountability for key abuse and malicious behavior, which makes it a promising technique for privacy-preserving user matching [41], [42]. Fig. 2 demonstrates a classic user matching scenario for mobile social networks.

In Fig. 2, anonymous bulletin boards allow senders and receivers to share information. Bilateral access control ensures that both parties are intended partners rather than malicious imposters upon correctly decrypting the ciphertext. Additionally, anonymity may be necessary to protect user privacy during the process. Message authenticity is necessary to prevent false or misleading information from being received due to ciphertext forgery. By providing bilateral access control, confidentiality and authenticity of messages, and identity privacy of sender and receiver, A-IB-ME can establish a positive social network environment. However, the full trust in the authority confronts the problems of key escrow and accountability for malicious behaviors in user matching. To further mitigate malicious behavior and enhance supervision and punishment mechanisms, accountability measures should be instituted when keys are abused. Overall, A-IB-ME provides all necessary functionalities to support possible solutions for user matching in mobile social networks, with specifical steps detailed below:

- *Step 1:* The authority executes the initialization algorithm $\text{Setup}(1^\lambda)$ after inputting the security parameter λ to obtain the master public key mpk and master secret key msk , then publishes mpk .
- *Step 2:* The sender with the identity ID_A interacts with the authority to obtain an encryption key $ek_{ID_A}(S)$ by the

encryption key generation algorithm $\text{EKGen}(msk, ID_A)$.

- *Step 3:* Similarly, to obtain the decryption key $dk_{ID_B}(S')$, the receiver with the identity ID_B interacts with the authority by the decryption key generation algorithm $\text{DKGen}(msk, ID_B)$.
- *Step 4:* When the sender ID_A wishes to connect with a potential partner ID_B , the sender encrypts his personal information using the encryption algorithm $\text{Enc}(M, ek_{ID_A}(S), S'', ID_B)$ after inputting the intended partner's identity ID_B . The resulting ciphertext is then stored on an anonymous bulletin board.
- *Step 5:* At a certain moment, the receiver ID_B wants to find a target partner. After inputting the identity of the target sender ID_A , if they are the intended partners, the receiver is able to decrypt the ciphertext stored on the anonymous bulletin board correctly by the decryption algorithm $\text{Dec}(CT_{ID_A, ID_B, S, S'}, dk_{ID_B}(S'), ID_A)$ and proceed with further communication.
- *Step 6:* If the anonymous bulletin board detects an encipher box $\mathcal{E}_{ID_A}$ from the user ID_A is maliciously propagated, the user ID_A can prove his innocence and the creator of the encipher box $\mathcal{E}_{ID_A}$ can be implicated by the tracing algorithm $\text{Trace}^1(\mathcal{E}_{ID_A}, ek_{ID_A}(S))$. Similarly, if the anonymous bulletin board detects a decoder box $\mathcal{D}_{ID_B}$ belonging to ID_B is unrightfully shared on the platform, its creator can be traced by the tracing algorithm $\text{Trace}^2(\mathcal{D}_{ID_B}, dk_{ID_B}(S'))$.

The aforementioned statement indicates that A-IB-ME is capable of facilitating privacy-preserving user matching in mobile social networks. Correspondingly, similar applications are also present in Tor [1].

IX. CONCLUSION

In this paper, we introduce the concept of A-IB-ME and present the formal syntax and security definition to mitigate the key escrow and achieve accountability for malicious behaviors. In A-IB-ME, when the encryption key and decryption key are generated, the sender and receiver engage in a fully-simulatable k -out-of- n OT protocol with the authority, which can ensure that the authority does not know which encryption key (decryption key) components the sender (or receiver) has. Then, if the encryption key and decryption key are abused to create an encipher box and a decoder box, the tracing algorithm can implicate its creator by observing the difference between the decryption or encryption capabilities of the decoder or encipher box and that of the decryption or encryption key. While implementing accountability for malicious behaviors, we also consider bilateral access control, message authenticity, and identity privacy of sender and receiver. Then we prove its privacy, authenticity, dishonest-sender security, dishonest-receiver security, and dishonest-authority security, which are reduced to CBDH, D-Lin, and DBDH assumptions without random oracles. Next, we evaluate the performance of A-IB-ME theoretically and experimentally, which validates its efficiency. Finally, we present how A-IB-ME can be applied to privacy-preserving user

matching in mobile social networks. In future work, we will consider A-IB-ME with the more efficient tracing algorithm.

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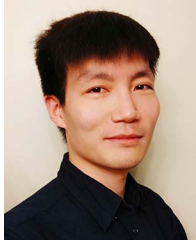
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